Ian Clowes

Michael Broderick

Elijah Dick

Brian Gardner 12/10/25

Final Project (Checkpoint 3):

Deliverables:

- Fully functional Tetris game that now includes complete GUI elements, full MVC structure, and the final gameplay features such as score, next-piece preview, game-over notification, and the updated Tetromino logic.
- Exception handling is now implemented to catch invalid moves and handle runtime issues cleanly while the game is running.
- Multithreading has been integrated through the main game loop thread so the piece-dropping logic runs smoothly without freezing the GUI.
- Generics were used where appropriate in the controller and model, mostly with collections for managing blocks and grid elements.
- Updated UML diagram reflecting the full design and final revisions. The UML PNG file can be found in the final project zip file, as well as the mp4 video showing the project running.

Reflection Questions:

- My main contributions during this final phase were finishing the UML updates and helping implement the exception handling and multithreading.
- An issue that we ran into that was a quick and easy fix was randomizing the next block after resetting the game. Previously, if the game was reset, the next block displayed would be the block that enters at the start of the next game.
- We continued to coordinate tasks via discord. This worked well for communication. However, after looking into other alternatives, I believe that git hub would have been a very reliable source. As it would have made collaborating on the code a lot easier.
- The main learning curve during this part of the project was how we could implement exception handling and multithreading into the project.
- If the project were to continue or be repeated, I think we would change to GitHub as our source of communication and collaboration.
- The provided resources were very effective in providing what was needed for the games GUI and other controls. It definitely would have been much harder to code without these libraries.

Division of Labor:
Ian: UML diagrams, checkpoint documentation, code review.
Elijah: Core game logic, block rotation logic, Tetromino behaviors, debugging.
Brian: Core game logic, colors, score/preview display, debugging.
Michael: UML diagrams, checkpoint documentation, code review